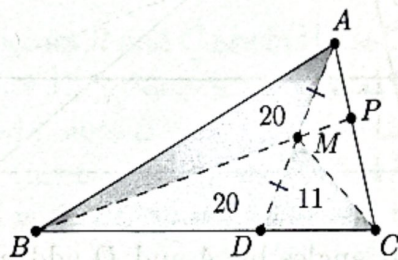


32. [AIME 2011] In triangle ABC , $11 \cdot AB = 20 \cdot AC$. The angle bisector of $\angle A$ intersects BC at point D , and point M is the midpoint of AD . Let P be the point of intersection of AC and BM . Find CP/PA .

First Solution. First, we recall the Angle Bisector Theorem and obtain

$$\frac{CD}{DB} = \frac{AC}{AB} = \frac{11}{20}.$$

We may scale up the triangle so that $[BDM] = 20$. Then we can use twice the Area Lemma (see Proposition 1.27) to find the areas $[CMD] = 11$ and $[AMB] = 20$.



The desired ratio is then found from yet another use of the Area Lemma as

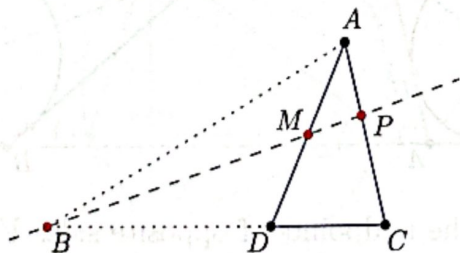
$$\frac{CP}{PA} = \frac{[CMB]}{[AMB]} = \frac{31}{20}.$$

Second Solution. After obtaining $CD/DB = 11/20$ as in the first solution we use Menelaus' Theorem for triangle ADC and line BM to get

$$\frac{AM}{MD} \cdot \frac{DB}{BC} \cdot \frac{CP}{PA} = 1.$$

As $AM/MD = 1$, this can be further rewritten as

$$\frac{CP}{PA} = \frac{BC}{DB} = \frac{CD}{DB} + 1 = \frac{31}{20}.$$



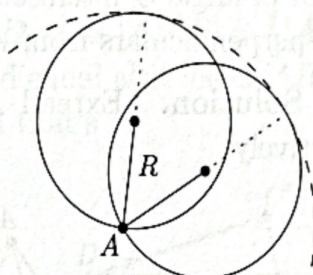
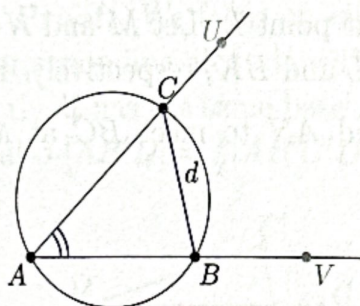
33. A variable segment BC of fixed length d moves such that its endpoints remain on the fixed rays AU , AV . Prove that the circumcircles of all possible triangles ABC are all tangent to a fixed circle.

Proof. The Law of Sines in triangle ABC implies that the circumradius R of triangle ABC is equal to

$$R = \frac{d}{2 \sin \angle BAC},$$

so in particular it is the same for all triangles ABC .

Since all these circumcircles pass through A , they are tangent to the circle with radius $2R$ centered at A , which is fixed.



34. Let ABC be a right triangle with $\angle A = 90^\circ$ and altitude AD . Let r, s, t be the inradii of triangles ABC, ADB , and ADC , respectively. Show that $r + s + t = AD$.

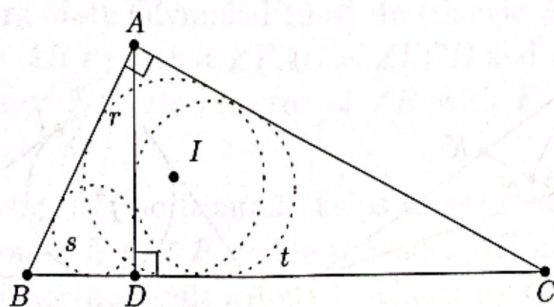
First Proof. We divide the desired equality by AD and note that triangles BAC, BDA , and ADC are pairwise similar (AA). We will use the proportionality of the triangles. The ratio s/AD in triangle BDA corresponds to r/AC in triangle ABC and likewise t/AD in triangle ADC corresponds to r/AB in triangle ABC . We are left to prove

$$\frac{r}{AD} + \frac{r}{AC} + \frac{r}{AB} = 1.$$

but if we denote the incenter of triangle ABC by I , the left-hand side is exactly

$$\frac{[BIC]}{[ABC]} + \frac{[CIA]}{[ABC]} + \frac{[AIB]}{[ABC]},$$

which clearly equals one.



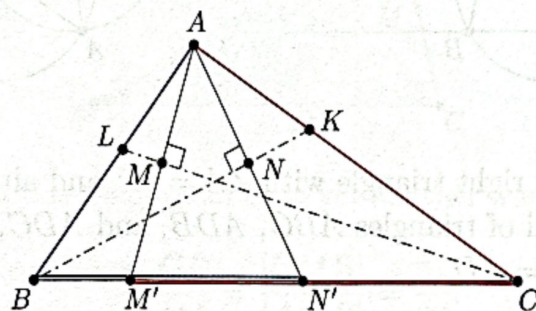
Second Proof. We recall that we have a formula for the inradius of a right triangle (see Proposition 1.16). Applying it three times, we obtain

$$r = \frac{AB + AC - BC}{2}, s = \frac{AD + BD - AB}{2}, t = \frac{AD + CD - AC}{2}.$$

After addition and some cancelling, we arrive at the result.

35. [AIME 2011] In triangle ABC , $BC = 125$, $CA = 120$, and $AB = 117$. The angle bisector of angle B intersects CA at point K , and the angle bisector of angle C intersects AB at point L . Let M and N be the feet of the perpendiculars from A to CL and BK , respectively. Find MN .

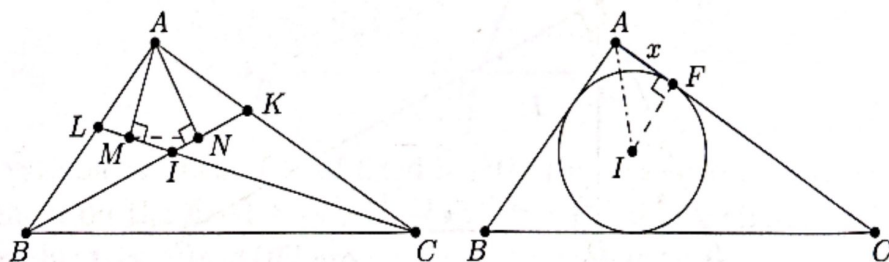
First Solution. Extend AM and AN to meet BC at M' and N' , respectively.



In triangle ACM' , the C -altitude and the angle bisector both coincide with CM , hence it is isosceles, $CM' = CA = 120$ and $AM' = 2 \cdot AM$. Analogously, $BN' = AB = 117$ and $AN' = 2 \cdot AN$. Line MN is therefore midline in triangle $AM'N'$ and $M'N' = 2 \cdot MN$. From $M'N' = BN' + M'C - BC$ we infer $MN = \frac{1}{2}(117 + 120 - 125) = 56$.

Second Solution. Observe that $I = BK \cap CL$ is the incenter of triangle ABC and recall the notorious angle $\angle BIC = 90^\circ + \frac{1}{2}\angle A$ (see Proposition 1.11). Also, $MINA$ is cyclic with diameter AI . Then by the Extended Law of Sines in triangle MIN we have

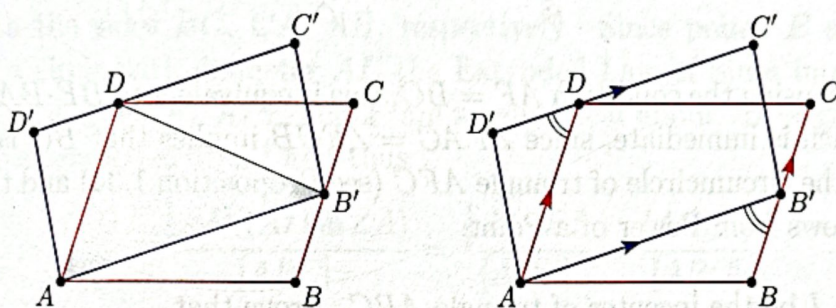
$$MN = AI \sin \left(90^\circ + \frac{1}{2}\angle A \right) = AI \cos \frac{1}{2}\angle A.$$



To give some meaning to the expression on the right-hand side we work with a separate diagram. We denote by F the point of contact of the incircle of triangle ABC with AC . Then from right triangle AIF we have $AF = AI \cos \frac{1}{2}\angle A$. Thus $MN = AF = x = \frac{1}{2}(b + c - a) = 56$ (see Proposition 1.15(a) if details are needed).

36. Let $ABCD$ and $AB'C'D'$ be parallelograms such that B' lies on the segment BC and D lies on the segment $C'D'$. Show that their areas are equal.

First Proof. We focus on triangle $AB'D$. It shares the base AD and the corresponding altitude with $ABCD$, thus $[AB'D] = \frac{1}{2}[ABCD]$. But similarly, it has the same base AB' and equal altitude as $AB'C'D'$ so we have also $[AB'D] = \frac{1}{2}[AB'C'D']$. And that's it!



Second Proof. We employ more computational approach. The desired equality $[ABCD] = [AB'C'D']$ rewrites as

$$AB \cdot AD \cdot \sin \angle BAD = AD' \cdot AB' \cdot \sin \angle B'AD'$$

or making use of parallel lines as

$$\frac{AB}{AB'} \cdot \sin \angle B'BA = \frac{AD'}{AD} \cdot \sin \angle AD'D.$$

But from the Law of Sines in triangle ABB' the left-hand side equals just $\sin \angle AB'B$ and likewise from the Law of Sines in triangle $AD'D$, the right-hand side is $\sin \angle D'DA$. Moreover, since $BC \parallel AD$ and $AB' \parallel C'D'$, the two angles are equal and we are done.

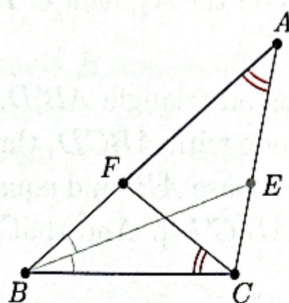
37. [St. Petersburg Math Olympiad 1994] In triangle ABC there is a point F on the side AB such that $\angle FAC = \angle FCB$ and $AF = BC$. Further, BE is the internal angle bisector of $\angle B$ with $E \in AC$. Show that $EF \parallel BC$.

Proof. Placing BC horizontally helps to see that we in fact need to prove that points E and F divide the sides AC and AB in the same ratios, since then the result follows by similarity of triangles ABC and

AFE . Thus, taking into account the Angle Bisector Theorem $CE/EA = BC/BA$, it suffices to prove that

$$\frac{BF}{FA} = \frac{BC}{BA}$$

and we may forget the segment BE .

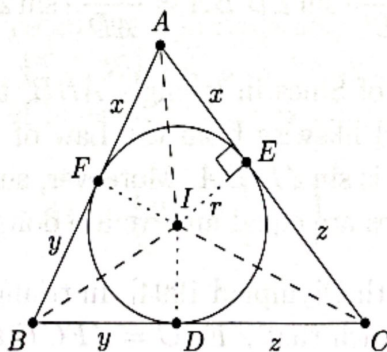


Now using the condition $AF = BC$, this is equivalent to $BF \cdot BA = BC^2$, which is immediate, since $\angle FAC = \angle FCB$ implies that BC is tangent to the circumcircle of triangle AFC (see Proposition 1.34) and the result follows from Power of a Point.

38. Let I be the incenter of triangle ABC . Prove that

$$\frac{AI^2}{bc} + \frac{BI^2}{ca} + \frac{CI^2}{ab} = 1.$$

First Proof. Let D, E, F be the points of contact of the incircle of triangle ABC with the sides BC, CA, AB , respectively.



From the right triangle IEA we have $IA^2 = r^2 + x^2$, where the inradius r can be found (see Proposition 1.26) as

$$r^2 = \frac{xyz}{x + y + z}.$$

Now, we can rewrite the desired equality only in terms of x , y , and z , which essentially solves the problem. Indeed,

$$\frac{AI^2}{bc} = \frac{\frac{xyz}{x+y+z} + x^2}{(x+y)(x+z)} = \frac{x(x^2 + xy + xz + yz)}{(x+y+z)(x+y)(x+z)} = \frac{x}{x+y+z},$$

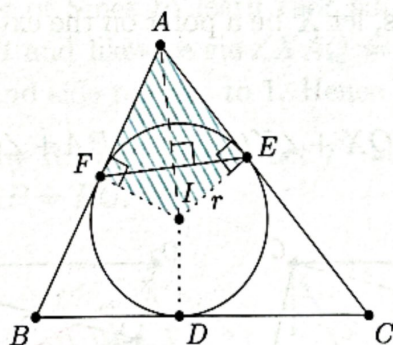
and the conclusion follows after we apply the same reasoning for distances BI and CI .

Second Proof. In order to give some meaning to the product in the denominator, we rewrite the first fraction as

$$\frac{AI^2}{bc} = \frac{\frac{1}{2} \cdot AI^2 \sin \angle A}{\frac{1}{2} bc \sin \angle A} = \frac{\frac{1}{2} AI \cdot (AI \sin \angle A)}{[ABC]}.$$

As in the first proof, let D , E , F be the points of contact of the incircle with the sides BC , CA , AB , respectively. Since points E and F lie on a circle with diameter AI , the Extended Law of Sines implies that $AI \sin \angle A = EF$. As E and F are symmetrical about AI , segment EF is perpendicular to AI and thus

$$\frac{\frac{1}{2} AI \cdot (AI \sin \angle A)}{[ABC]} = \frac{\frac{1}{2} AI \cdot EF}{[ABC]} = \frac{[AEIF]}{[ABC]}.$$



Likewise we obtain

$$\frac{BI^2}{ac} = \frac{[BFID]}{[ABC]} \quad \text{and} \quad \frac{CI^2}{ab} = \frac{[CDIE]}{[ABC]}$$

and the result follows.

Third Proof. We recall that the whole left-hand side can be expressed in terms of the triangle sides. Combining the formula for the length of the angle bisector (see Corollary 1.24) with the known ratio in which it is divided by I (see Corollary 1.28) gives

$$AI^2 = \left(\frac{b+c}{a+b+c} \right)^2 \cdot bc \cdot \left(1 - \left(\frac{a}{b+c} \right)^2 \right),$$

from which we after some manipulation find

$$\frac{AI^2}{bc} = \frac{(b+c-a)(a+b+c)}{(a+b+c)^2} = \frac{b+c-a}{a+b+c}.$$

Again, the result follows after applying the same to BI and CI .

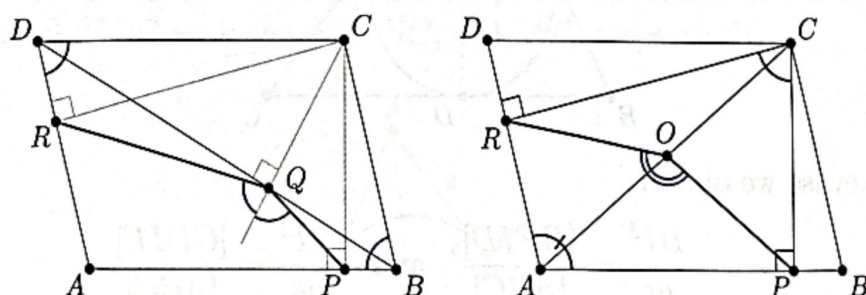
Remark. This problem has the following generalization which can be proved using the idea from the second presented proof. If P and Q are isogonal conjugates with respect to triangle ABC and both lie in its interior, then

$$\frac{AP \cdot AQ}{bc} + \frac{BP \cdot BQ}{ca} + \frac{CP \cdot CQ}{ab} = 1.$$

39. In parallelogram $ABCD$ with $\angle BAD > 90^\circ$, show that the circle passing through the projections of C onto AB , BD , and DA , respectively, passes through the center of the parallelogram.

Proof. Denote the feet of projections onto AB , BD , DA by P , Q , R , respectively, and observe that since $\angle CPB = \angle CQB = 90^\circ$, points C , Q , P , B are concyclic and likewise C , Q , R , D are concyclic. For notational purposes, let X be a point on the extension of CQ beyond Q . We angle-chase:

$$\angle PQR = \angle PQX + \angle XQR = \angle CBA + \angle CDA = 2 \cdot \angle CBA.$$



Next, we erase Q and aim to find $\angle POR$ where O is the center of the parallelogram. Since $\angle APC = \angle ARC = 90^\circ$, points A , P , C , R lie on a single circle and O (being the midpoint of its diameter AC) is its center. Hence

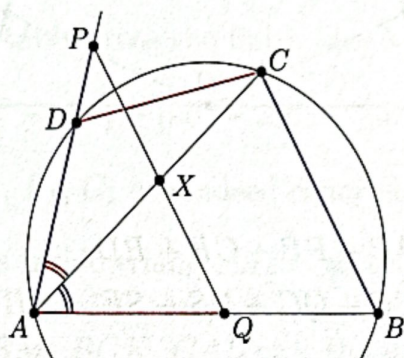
$$\angle POR = 2 \cdot \angle PCR = 2 \cdot (180^\circ - \angle DAB) = 2 \cdot \angle CBA$$

and the proof is complete.

40. Let $ABCD$ be a cyclic quadrilateral. Let P be the point on the ray AD such that $AP = BC$ and let Q be the point on the ray AB such that $AQ = CD$. Prove that the line AC cuts PQ at its midpoint.

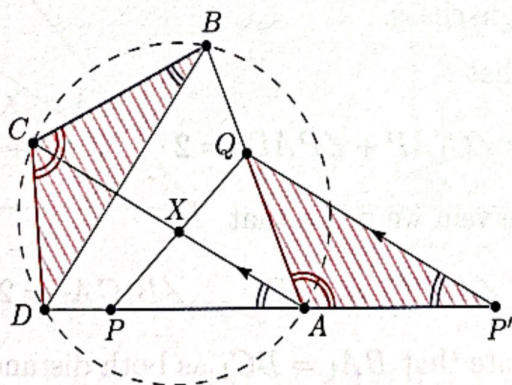
First Proof. The Law of Sines should be the first idea. Denote the intersection of AC and PQ by X and use the Ratio Lemma (see Proposition 1.18) for triangle PAQ to obtain

$$\frac{PX}{XQ} = \frac{AP}{AQ} \cdot \frac{\sin \angle PAX}{\sin \angle XAQ}.$$



Denoting the radius of the circumcircle of $ABCD$ by R we can use the Extended Law of Sines to learn that $\sin \angle PAX = \sin \angle DAC = CD/2R = AQ/2R$ and likewise $\sin \angle XAQ = AP/2R$. Plugging it in, the whole right-hand side reduces to 1. Hence $PX = XQ$ as desired.

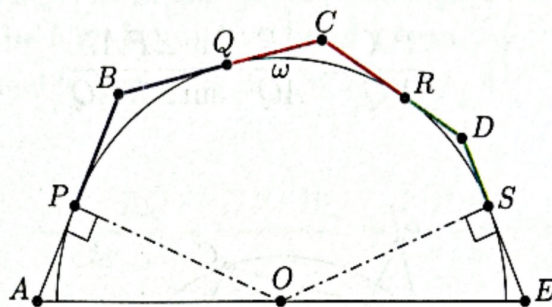
Second Proof. (by Richard Stong) Let P' be the other point on line AD with $AP' = AP = BC$.



Then as $ABCD$ is cyclic, we have $\angle QAP' = \angle BCD$ and triangles QAP' and DCB are congruent (SAS). Hence $\angle QP'A = \angle DBC = \angle DAC$ and therefore $P'Q$ is parallel to AC . Now in triangle $\triangle PP'Q$, A is the midpoint of PP' and AC is parallel to $P'Q$. Hence AC is the midline and in particular $X = AC \cap PQ$ is the midpoint of PQ .

41. [Junior Balkan 2009] Let $ABCDE$ be a convex pentagon such that $AB + CD = BC + DE$ and a circle ω with center O on the side AE is tangent to the sides AB, BC, CD and DE at points P, Q, R and S , respectively. Prove that the lines PS and AE are parallel.

Proof. First, we use Equal Tangents to transform the given metric condition into something more approachable.



We have

$$AB + CD = AP + PB + CR + RD,$$

$$BC + DE = BQ + QC + DS + SE = PB + CR + RD + SE,$$

thus by comparison, we obtain $AP = SE$. Now, the right triangles AOP and EOS are congruent (SAS) and so they have equal corresponding altitudes. Then the points P and S have the same distance from the line AE (and lie in the same half-plane) implying the desired $PS \parallel AE$.

42. Let P be a point inside acute-angled triangle ABC with $\angle BPC = 180^\circ - \angle A$. Denote by A_1, B_1, C_1 its reflections over the sides BC, CA, AB , respectively. Prove that the points A, A_1, B_1, C_1 are concyclic.

Proof. We angle-chase.

First, observe that

$$\angle C_1AB_1 = \angle C_1AP + \angle PAB_1 = 2 \cdot (\angle BAP + \angle PAC) = 2\angle A$$

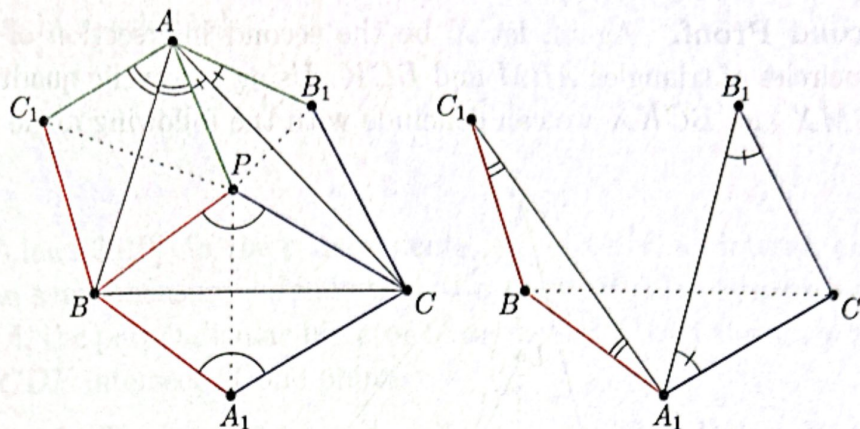
and in the same vein we prove that

$$\angle A_1BC_1 = 2\angle B, \quad \angle B_1CA_1 = 2\angle C.$$

Moreover, we note that $BA_1 = BC_1$ as both distances are by symmetry equal to BP and similar reasoning shows $CA_1 = CB_1$. Keeping in mind we only need to find angle $B_1A_1C_1$ and that we know $\angle CA_1B = \angle BPC = 180^\circ - \angle A$, we lose no information if we erase points P and A .

Now we clearly see the isosceles triangles A_1BC_1 and B_1CA_1 , which give us

$$\angle C_1A_1B = 90^\circ - \angle B \quad \text{and} \quad \angle CA_1B_1 = 90^\circ - \angle C$$



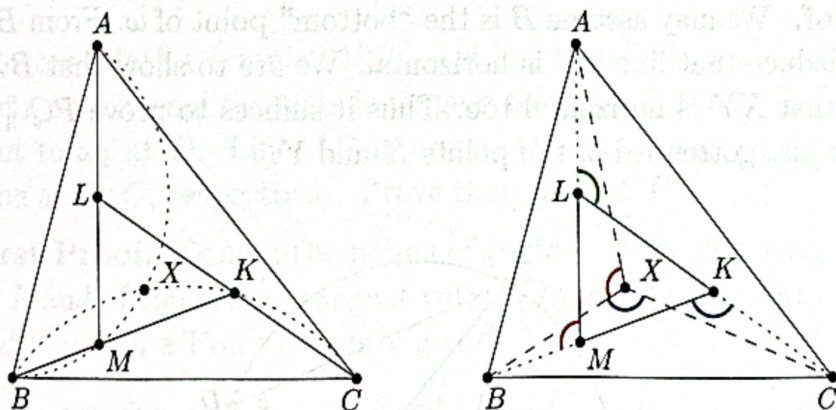
and we can comfortably cross the finish line:

$$\angle B_1 A_1 C_1 = (180^\circ - \angle A) - (90^\circ - \angle B) - (90^\circ - \angle C) = 180^\circ - 2\angle A.$$

Then points A, A_1, B_1, C_1 are indeed concyclic.

43. Triangle KLM lies inside triangle ABC so that points K, L, M lie on the segments CL, AM, BK , respectively. Prove that the circumcircles of the triangles ABM, BCK, CAL pass through a common point.

First Proof. Let X be the second intersection of the circumcircles of triangles ABM and BCK . We will focus on external angles in triangle KLM , whose sum is 360° .



Since $ABMX$ and $BCKX$ are cyclic, we have

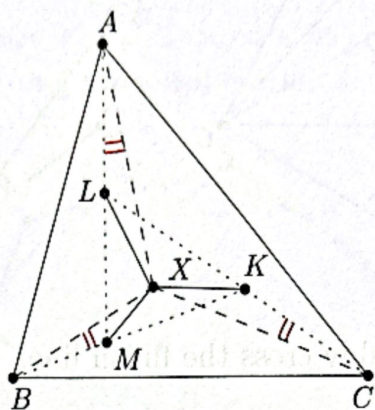
$$\angle BKC = \angle BXC, \quad \angle AMB = \angle AXB,$$

so we can find angle CXA as

$$\begin{aligned} \angle CXA &= 360^\circ - \angle BXC - \angle AXB = 360^\circ - \angle BKC - \angle AMB \\ &= \angle CLA. \end{aligned}$$

Thus $CALX$ is cyclic and we are done.

Second Proof. Again, let X be the second intersection of the circumcircles of triangles ABM and BCK . Using the cyclic quadrilaterals $ABMX$ and $BCKX$ we can conclude with the following angle chase

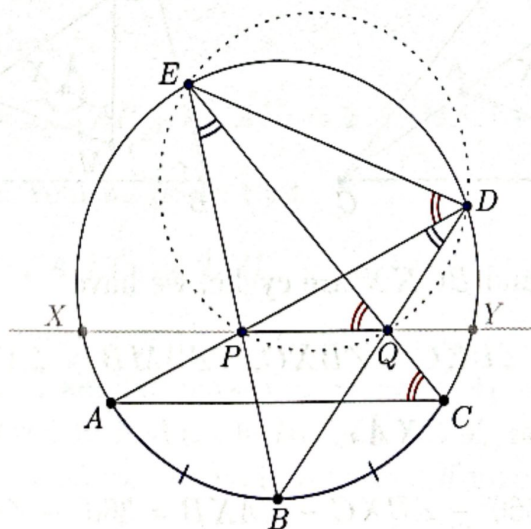


$$\angle LCX \equiv \angle KCX = \angle KBX \equiv \angle MBX = \angle MAX \equiv \angle LAX.$$

Remark. Did you notice that this is just Miquel's pivot theorem (Theorem 1.33) in disguise?

44. Let the pentagon $ABCDE$ inscribed in circle ω satisfy $BA = BC$. The line joining $P = BE \cap AD$ and $Q = CE \cap BD$ intersects ω at points X , Y . Prove that $BX = BY$.

Proof. We may assume B is the "bottom" point of ω . From $BA = BC$ we deduce that line AC is horizontal. We are to show that $BX = BY$, i.e. that XY is horizontal too. Thus it suffices to prove $PQ \parallel AC$. We have just gotten rid of the points X and Y .



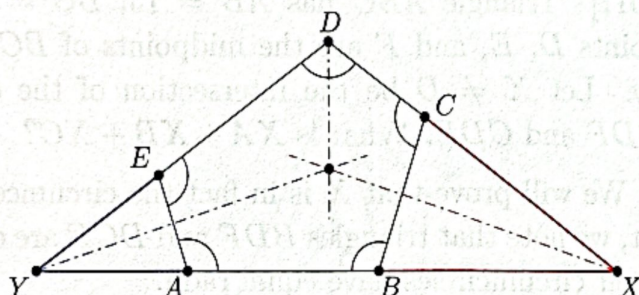
Since angles ADB and BEC subtend equal arcs, they are equal themselves. Thus, $PQDE$ is cyclic. Focusing on angles by the line CE we

obtain the desired

$$\angle EQP = \angle EDP \equiv \angle EDA = \angle ECA.$$

45. [Poland 2010] In the convex pentagon $ABCDE$ all interior angles have the same measure. Prove that the perpendicular bisector of segment EA , the perpendicular bisector of segment BC and the angle bisector of $\angle CDE$ intersect at one point.

Proof. The trick is to extend AB to meet CD , DE at X , Y , respectively. The triangles BCX and AEY are then isosceles so the perpendicular bisector of EA is the same line as the angle bisector of angle DYX and likewise, the perpendicular bisector of BC coincides with the angle bisector of angle YXD . The three lines thus concur at the incenter of triangle YXD .



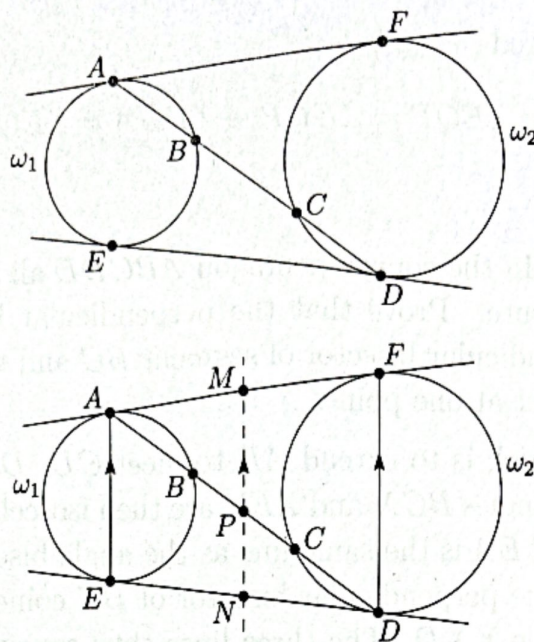
46. [Moscow Math Olympiad 1970] Let ω_1, ω_2 be two circles. One of their common external tangents is tangent to ω_1 at A , the second one is tangent to ω_2 at D . Line AD intersects the circles ω_1, ω_2 for the second time at B, C , respectively. Prove that $AB = CD$.

First Proof. Denote the points of contact of the first tangent with ω_2 by F and of the second tangent with ω_1 by E . By symmetry, $ED = AF$ and Power of a Point instantly gives

$$AC \cdot AD = p(A, \omega_2) = AF^2 = ED^2 = p(D, \omega_1) = BD \cdot AD.$$

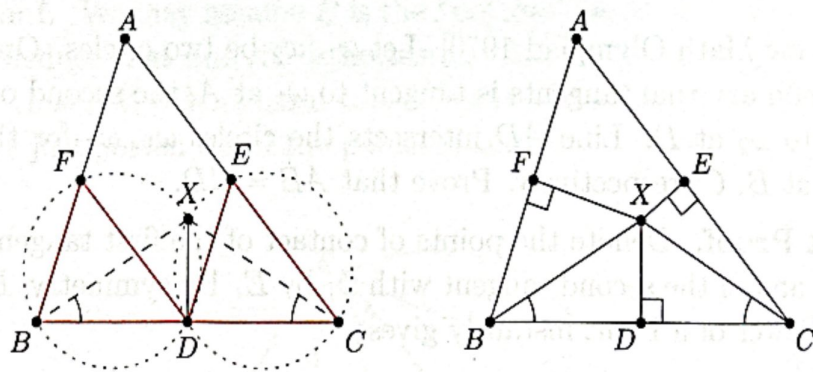
Cancelling AD we get $AC = BD$. Hence $AB = CD$.

Second Proof. We offer another approach. Define E, F as above and let P be the midpoint of AD . If we prove that P lies on the radical axis of ω_1, ω_2 , we are done, since from $PA \cdot PB = PC \cdot PD$ and $PA = PD$ we easily conclude $PB = PC$ and $AB = CD$. Since the midpoints M, N of AF, DE , respectively, lie on the radical axis (indeed, $MA^2 = MF^2$ and $NE^2 = ND^2$), it is enough to prove M, N, P collinear. But that's immediate as all three points lie on the midline of trapezoid $AEDF$.



47. [AMC12 2011] Triangle ABC has $AB = 13$, $BC = 14$, and $CA = 15$. The points D , E , and F are the midpoints of BC , CA , and AB , respectively. Let $X \neq D$ be the intersection of the circumcircles of triangles BDF and CDE . What is $XA + XB + XC$?

Solution. We will prove that X is in fact the circumcenter of triangle ABC . First, we note that triangles BDF and DCE are congruent (SSS) and thus their circumcircles have equal radii.



Hence the inscribed angles which correspond to the common chord XD are the same in both circles. Thus $\angle DBX = \angle XCD$, implying that triangle BCX is isosceles. Moreover, XD as a median in an isosceles triangle is perpendicular to the base BC . Thus, XB and XC are diameters of the respective circles, which also implies $\angle BFX = 90^\circ$ and $\angle XEC = 90^\circ$. But then DX , EX , and FX are perpendicular bisectors in triangle ABC , so their intersection X is indeed the circumcenter of triangle ABC .

Denoting the circumradius of triangle ABC by R , we are asked to find

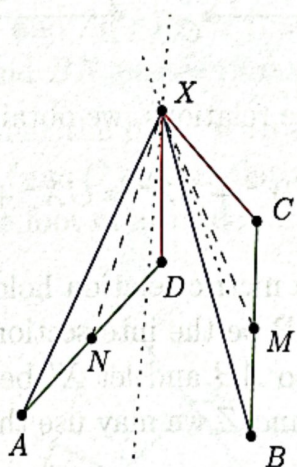
3R. We recall the xyz formula for R (see Proposition 1.26)

$$R = \frac{(x+y)(y+z)(z+x)}{4\sqrt{xyz(x+y+z)}},$$

and plug in $x = 7$, $y = 6$, and $z = 8$. The result is $3R = 195/8$.

48. Let $ABCD$ be a quadrilateral with segments BC and AD equal and AB not parallel to CD . Denote by M , N the midpoints of BC and AD , respectively. Prove that the perpendicular bisectors of AB , MN , and CD pass through a common point.

Proof. Two important ideas lie behind the following short solution. First, the perpendicular bisector is just a locus of equidistant points, and second, if we are proving concurrence of three lines, we often start by intersecting two of them.



Let X be the intersection of the perpendicular bisectors of AB and CD . Moreover, it suffices to keep in mind just $XA = XB$ and $XC = XD$ and not even bother drawing the perpendicular bisectors. But now triangles XBC and XAD are congruent (SSS)! It follows that the corresponding medians XM and XN are also equal, or in other words, that X lies on the perpendicular bisector of MN .

49. Carnot's³ Theorem.

Let X , Y , and Z lie on the sides BC , CA , AB , respectively, of a triangle ABC . Show that the perpendiculars from X , Y , Z to the respective triangle sides meet at one point if and only if

$$BX^2 + CY^2 + AZ^2 = CX^2 + AY^2 + BZ^2.$$

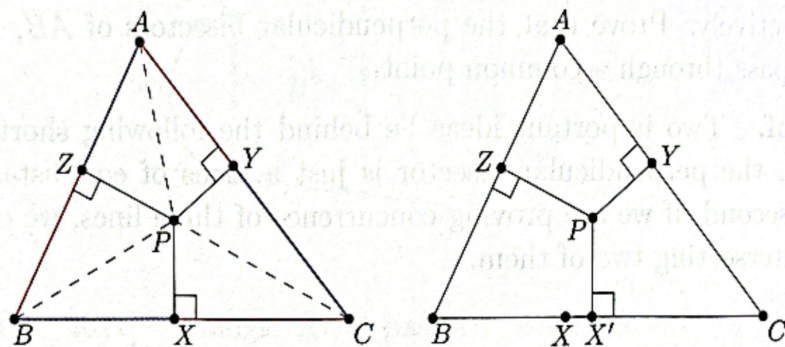
³Lazare Nicolas Marguerite Carnot (1753–1823) was an amateur mathematician and French minister of war during the French revolutionary wars.

Proof. First, assume the perpendiculars are concurrent at P . From perpendicularity criterion (see Proposition 1.22) for $PX \perp BC$, we learn

$$BP^2 - PC^2 = BX^2 - CX^2$$

and analogously we find

$$CP^2 - PA^2 = CY^2 - AY^2 \quad \text{and} \quad AP^2 - PB^2 = AZ^2 - BZ^2.$$



Now if we add the three relations, we obtain the desired

$$BX^2 + CY^2 + AZ^2 = CX^2 + AY^2 + BZ^2.$$

Conversely, assume the metric relation holds for some X , Y , and Z on the triangle sides. Let P be the intersection of the perpendiculars from Y to AC and from Z to AB and let X' be the projection of P to BC . Now for points X' , Y , and Z we may use the first part of this statement and obtain

$$BX'^2 + CY^2 + AZ^2 = CX'^2 + AY^2 + BZ^2,$$

which after comparison with the given condition yields

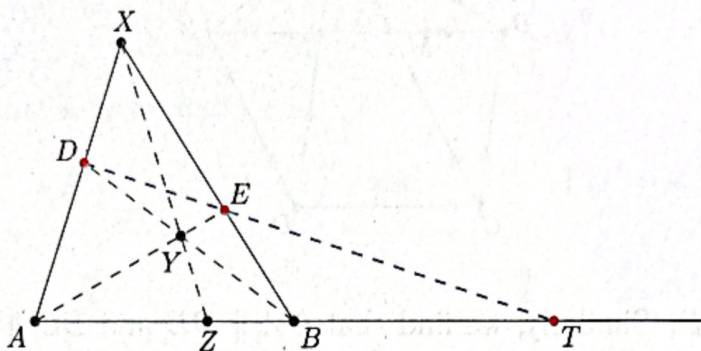
$$BX'^2 - CX'^2 = BX^2 - CX^2.$$

We claim this can only happen if $X = X'$. Indeed, should we have $BX < BX'$, then $CX > CX'$ and the left-hand side has greater value. The case $BX > BX'$ is treated in the same fashion. We may now conclude.

50. [South Africa 2003] In a given pentagon $ABCDE$, triangles ABC , BCD , CDE , DEA and EAB all have the same area. The lines AC and AD intersect BE at points M and N . Prove that $BM = EN$.

Proof. The equality $[BCD] = [CDE]$ means that as the triangles have common base CD , they also have the same altitude. In other words

(which are not parallel, as $\angle A \neq \angle B$, if in doubt see Example 1.20) and aim to compute the ratio AT/TB .



The correct technique here is to compare Ceva's Theorem for cevians passing through Y with Menelaus' Theorem for the collinear points D, E, T , both with respect to triangle ABX . We obtain

$$\frac{AZ}{ZB} \cdot \frac{BE}{EX} \cdot \frac{XD}{DA} = 1 \quad \text{and} \quad \frac{AT}{TB} \cdot \frac{BE}{EX} \cdot \frac{XD}{DA} = 1.$$

Comparing the two gives

$$\frac{AZ}{ZB} = \frac{AT}{TB},$$

which shows that the ratio AT/TB is independent of the positions of X and Y . Moreover, since T lies outside the segment AB , this ratio determines it uniquely. Hence all the lines DE pass through T .

53. Let ω_1 and ω_2 be two circles centered at distinct points O_1 and O_2 and with radii r_1, r_2 , respectively.

- Find the locus of points X for which $p(X, \omega_1) - p(X, \omega_2)$ is constant.
- Find the locus of points X for which $p(X, \omega_1) + p(X, \omega_2)$ is constant.

Solution.

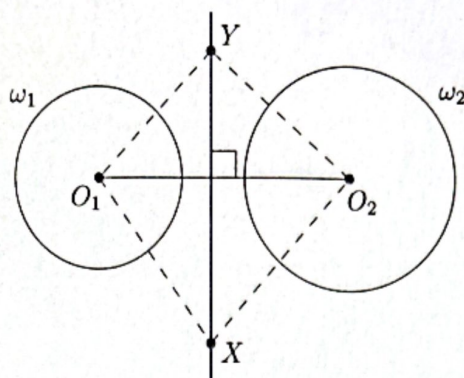
- Assume for some two points X and Y , we have

$$p(X, \omega_1) - p(X, \omega_2) = p(Y, \omega_1) - p(Y, \omega_2).$$

From the very definition of Power of a Point we obtain $p(X, \omega_1) = O_1X^2 - r_1^2$ and similarly we rewrite the remaining three terms. After some cancellation we get

$$O_1X^2 - O_2X^2 = O_1Y^2 - O_2Y^2,$$

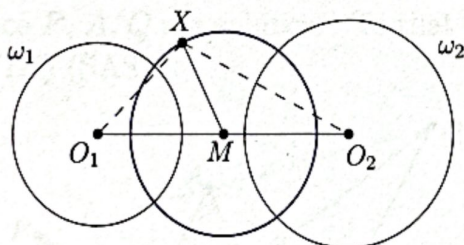
which means (by Proposition 1.22) that $XY \perp O_1O_2$. In other words, we see that the desired points must lie on a line perpendicular to O_1O_2 . By reversing the previous calculation, we see that all such points satisfy the condition.



- (b) Again we make use of the definition of Power of a Point. After writing $p(X, \omega_1) + p(X, \omega_2) = XO_1^2 - r_1^2 + XO_2^2 - r_2^2$ we see that in fact we need $XO_1^2 + XO_2^2$ to be constant. Now the trick is to look at the midpoint M of O_1O_2 . By the median formula (see Proposition 1.24) we have

$$XM^2 = \frac{1}{2}(XO_1^2 + XO_2^2) - \frac{1}{4}O_1O_2^2,$$

which means that the desired locus is formed by points with constant distance XM . In other words, it is a circle centered at M . The case when the triangle XO_1O_2 degenerates into a line is treated in the same vein and the resulting points complete the circle.



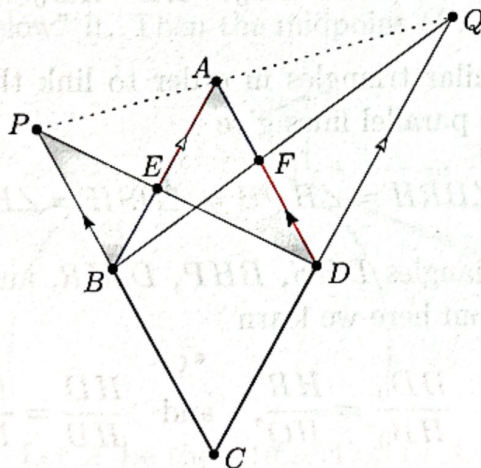
Chapter 5

Solutions to Advanced Problems

1. [Romania 2004] On the sides AB and AD of the rhombus $ABCD$ consider the points E and F such that $AE = DF$. Let $BC \cap DE = P$ and $CD \cap BF = Q$. Prove that points P , A , and Q are collinear.

Proof. Pairs of parallel lines and equal segments suggest approaching the problem in terms of ratios.

Observe $PB \parallel AD$ and $BA \parallel DQ$. If we prove that the triangles PBA and ADQ are similar, then the corresponding sides PA and AQ are also parallel and hence P , A , Q are collinear. To that end it suffices to prove $PB/AD = BA/DQ$ (SAS).



This is easily accomplished once we realize that $\triangle EPB \sim \triangle EDA$ (AA) and likewise $\triangle FQD \sim \triangle FBA$, since then

$$\frac{PB}{AD} = \frac{BE}{EA} = \frac{AF}{FD} = \frac{BA}{DQ}.$$

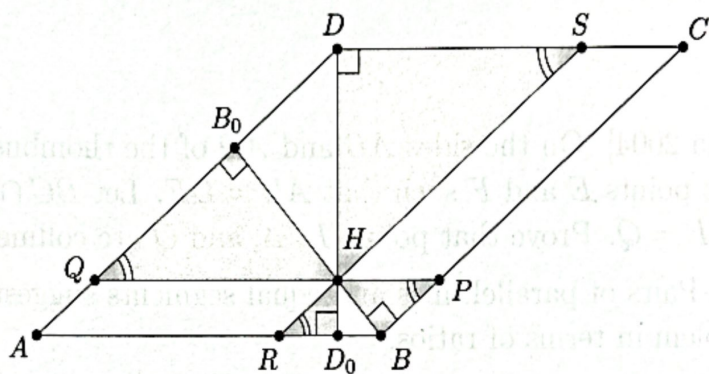
2. [Switzerland 2011] Let $ABCD$ be a parallelogram such that the triangle ABD is acute and has orthocenter H . The line through H parallel to AB cuts AD and BC at Q and P , respectively, while the line through H parallel to BC cuts AB and CD at R and S , respectively. Prove that the points P, Q, R, S lie on the same circle.

Proof. We choose to define the orthocenter as the intersection of altitudes DD_0 and BB_0 , also we exclude the diagonal BD from our diagram.

We aim to prove concyclicity in the language of ratios. Namely, by proving

$$HQ \cdot HP = HS \cdot HR,$$

which suffices by Power of a Point.



Also, we observe that quadrilateral DB_0D_0B is cyclic (as $\angle BB_0D = 90^\circ = \angle DD_0B$) so by Power of a Point we have

$$HB \cdot HB_0 = HD \cdot HD_0. \quad (*)$$

Now we find similar triangles in order to link the two alike relations. Indeed, since the parallel lines give

$$\angle BRH = \angle HPB = \angle DSH = \angle HQB_0,$$

the four right triangles DHS , BHP , D_0HR , and B_0HQ are pairwise similar (AA). From here we learn

$$\frac{HD_0}{HB_0} = \frac{HR}{HQ}, \quad \text{and} \quad \frac{HD}{HB} = \frac{HS}{HP}$$

and after multiplying the two and comparing with $(*)$, we obtain the result.

3. [Baltic Way 2010] Let ABC be an acute-angled triangle. Let D and E be points on the sides AB and AC such that B, C, D , and E lie on the same circle. Further, suppose the circle through D, E , and A intersects